

Geometry

Unit 8

Lesson 7 Homework

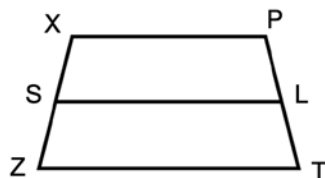
Name _____

Date _____

Period _____

$XZTP$ is a trapezoid where $\overline{XP} \parallel \overline{ZT}$. $\overline{SL}$ is a midsegment of $XZTP$ and $XS = 7$, $LT = 6$, $SL = 10$, $TZ = 2x - 4$, and $XP = x - 3$.

1. What is the relationship of $\overline{SL}$ to $\overline{XP}$ and $\overline{ZT}$?



2. What is the value of x ?

3. What is the length of $\overline{XP}$ and $\overline{ZT}$?

Trapezoid $BHLN$ is shown where $\overline{BH} \parallel \overline{NL}$ with midsegment $\overline{DP}$ and $BH = 5x + 12$, $NL = 3x$, and $DP = 5x$.

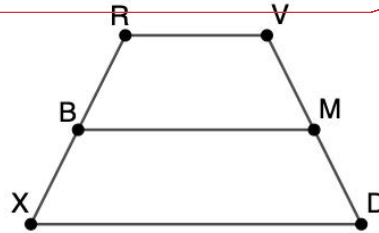
4. Find the value of x .



5. Find the length of $\overline{DP}$.

6. Find the lengths of $\overline{BH}$ and $\overline{NL}$.

Isosceles trapezoid $RVDX$ is shown with midsegment $\overline{BM}$ where $RV = 2x + 1$, $XD = 3x + 2$, $BM = 2x + 4$ and $MD = 3$.



Commented [DB1]: <https://www.geogebra.org/calculator/vcppzgw3>

7. What is the value of x ?

8. What is the length of $\overline{BM}$?

9. What is the length of $\overline{XD}$ and $\overline{RV}$?

10. What is the perimeter of $RVDZ$?